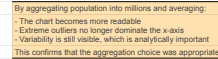


Overall conclusion

When the urban share increases, the rural share decreases

This confirms the logical relationship:
Urban share + Rural share = 100%

This validates that the data and transformations were done correctly.



As population size increases:

- Urban and rural shares fluctuate significantly
 - There is no clear upward or downward trend
 - Countries with similar population sizes can have very different urbanization levels
- Population size alone does not determine how urbanized a country is.

Across small, medium, and large population groups:

- Some countries are highly urbanized
 - Others remain predominantly rural
 - This variability persists even among very large populations
- Urbanization depends more on economic structure, development

Overall conclusion

The visualization shows an inverse relationship between urban and rural population shares, as expected. However, no clear trend emerges between national population size and the degree of urbanization. Countries with similar population sizes can have very different urban-rural distributions, indicating that population size alone does not explain urbanization patterns.

Investigating access by area

	was bas r	was lin r	was usmp r	was usmp r	was bas r	was usmp r	was usmp r	was bas r	was lin r	was usmp r	was usmp r	was usmp r
MAX	37.42299287	33.591371	30.3697938	100	42.16438068	41.21598167	40.81512342	100	34.27778069	19.8358081	36.3210565	
73.22042625				100	91.76276184	91.76276184	90	48.68186495				
MIN	39.4647478	3.85652635	4.41910054	1.91488504	1.33591865	5.83656254	8.72226144	1.22893735	3.381398545	1.71787804	0.31311484	
MEAN	37.3841397	4.07292181	0.635647404	0.173761683	1.81230365	3.22047913	4.69606289	98.10662849	0.05	0.3948740029		
MODE	100	0	0	0	0	0	0	0	0	0	0	
STDEV	85.8433106	0	0.03260226211	64.8242611	0	0.1648104016	90.562371	0	0	0	0	
1ST Quartile	99.87521524	4.85121957	0.273090022	1.876190393	99.11024429	8.553171212	13.266393	1.57471481	99.9418103	3.966347184	0.259901302	0.15976656
2ND Quartile	14.24221428	4.85121957	0.254403699	1.876190393	34.29521319	8.553171212	13.1702836	1.57471481	7.33654051	3.966347184	0.259901302	0.15976656
3RD Quartile	100	0	0	0	0	0	0	0	0	0	0	
STANDARD	0.08887616	0.96048653	0.714477036	3.923975931	21.50665741	8.704766548	11.82849434	8.847951385	1.58587611	6.53506705	3.2545732	0.869631624

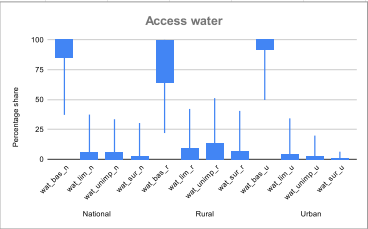
We observe that:
Mean $\neq$ Median $\neq$ Mode for all four features
this tells us that:
The distributions are not normal (not symmetric).
This is expected in global data because:
The countries are very heterogeneous
The access levels are clustered at extremes (near 0% or near 100%)

Interpretations:

National share of population with at least basic drinking water access

Most countries have very high national access to basic drinking water, and many have full coverage. However, a small group of countries with very low access creates inequality at the global level. This is why the mean is lower than the median and why variability exists despite high overall access.

National share of population with limited drinking water service	wat_lim_n	In most countries, limited water access is very low or zero, which is why the median is close to 0 and the most common value is 0. However, a small group of countries still has significant limited access, raising the mean and increasing the spread.
National share of population with unimproved drinking water service	wat_unimp_n	Most countries have unimproved water access that is very low or near zero, which is why the median is small and the mode is 0. However, a smaller group of countries still has high unimproved access, pushing the mean upward and creating noticeable variability across the dataset.
National share of population using surface water for drinking water	wat_sur_n	Surface water use is zero for most countries, which is why the median and mode are 0 and the IQR is small. However, a few countries still rely heavily on surface water, which increases the maximum and makes the standard deviation larger than what the median alone suggests.
Rural share of population with at least basic drinking water access	wat_bas_r	Most countries have high rural basic water access and many are near 100%. However, a minority of countries have very low rural access, which creates strong inequality and explains the large IQR and standard deviation.
Rural share of population with limited drinking water service	wat_lim_r	Most countries have very low (often zero) rural limited access, but a minority of countries still has high limited service in rural areas, which raises the mean and increases variability.
Rural share of population with unimproved drinking water service	wat_unimp_r	Most countries have low rural unimproved access, but a minority of countries still has very high unimproved water use in rural areas. This is why the mean is higher than the median and why IQR/STD are relatively large.
Rural share of population using surface water for drinking water	wat_sur_r	Most countries have very low (often near zero) rural surface water use, but a few countries rely heavily on surface water in rural areas, which raises the maximum and increases variability.
Urban share of population with at least basic drinking water access	wat_bas_u	Urban basic water access is very high for most countries and often reaches 100%. The distribution is tight (small IQR and STD), meaning urban access is relatively similar across countries, with a few low-access exceptions.
Urban share of population with limited drinking water service	wat_lim_u	Urban limited access is usually very low or zero, but a few countries still show high limited service in urban areas, raising the maximum and increasing spread.
Urban share of population with unimproved drinking water service	wat_unimp_u	Urban unimproved access is near zero for most countries, but a few countries still have noticeable unimproved access, which increases the maximum and spread.
Urban share of population using surface water for drinking water	wat_sur_u	Urban surface water use is essentially zero for most countries (median mode = 0, tiny IQR). Only a small number of countries have noticeable urban surface water reliance, which explains the maximum and the standard deviation.

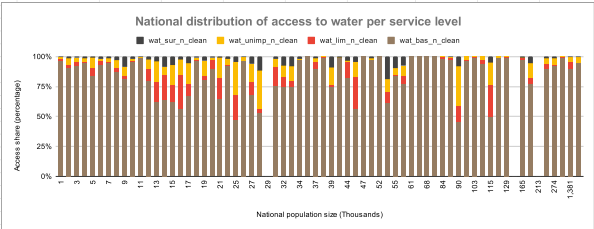


Interpretation by service type			
At least basic access	National (wat_bas_n)	box high (near the top) → most countries have high national basic access, but some lower outliers exist.	Urban > National > Rural in basic access, and rural has the largest inequality
	Rural (wat_bas_r)	box lower and much wider → rural basic access varies a lot; some countries are far behind.	
	Urban (wat_bas_u)	box highest and tightest → urban basic access is near universal in most countries.	
Limited access	Urban (wat_lim_u)	box close to 0 → in most countries, urban limited access is almost zero.	limited access is generally low, but it appears more in rural areas and is concentrated in certain countries.
	National (wat_lim_n)	also close to 0 but slightly higher spread than urban.	
	Rural (wat_lim_r)	higher box and longer whiskers → a subset of countries still has meaningful rural limited access.	
Unimproved access	Urban (wat_unimp_u)	box very close to 0 → unimproved access is rare in urban areas for most countries.	unimproved access is mostly a rural issue, with strong cross-country differences.
	National (wat_unimp_n)	low but slightly higher than urban.	
	Rural (wat_unimp_r)	clearly higher box and long upper whisker → rural unimproved access is a major problem in some countries.	
Surface water	Urban (wat_sur_u)	almost all values near 0 → urban surface water use is nearly nonexistent.	surface water use is generally rare, but when it exists, it is mostly rural and concentrated in a few countries.
	National (wat_sur_n) Rural (wat_sur_r)	Rural 0 but with some countries higher. higher and more spread → some countries rely heavily on surface water in rural areas.	

Overall pattern across areas	
Urban is best and most consistent	For wat_bas_u, the box is very high and relatively tight → most countries have very high urban basic access with limited variation.
Rural shows the biggest inequality	For wat_bas_r, the box is lower and much taller than urban → rural basic access is lower and much more unequal across countries. For "problem" indicators (wat_lim_r, wat_unimp_r, wat_sur_r), rural boxes and whiskers are generally higher → rural areas have more limited, unimproved, and surface water use.
National sits between Urban and Rural	National distributions usually fall between urban and rural, which makes sense because national values combine both.

The box-and-whisker plots show clear differences by area. Urban basic water access is consistently high across countries, with a tight distribution close to 100%. Rural basic access is lower and much more variable, indicating stronger inequality between countries. For limited, unimproved, and surface water indicators, distributions are concentrated near zero in urban areas but are higher and more dispersed in rural areas, meaning these problems are primarily rural and concentrated in a subset of countries.

Investigating access by population size



- Most population-size bins will be dominated by basic access (large blue section).
- The remaining sections (limited/unimproved/surface) appear as smaller bands.
- We don't see a strong "population size → access" pattern; differences are more influenced by development/income and rural/urban structure.